

NAME: Diya Shah

SERVER ASSESSMENT

Security and Maintenance

Section A: Scenario-Based Concept Application

1. Core Services: Install Windows Server and promote it to a Domain Controller with AD DS and DNS.

To configure the server, Windows Server 2022 is first installed on a virtual machine using VMware or VirtualBox. After installation, a static IP address is assigned to the server. Using Server Manager, the Active Directory Domain Services (AD DS) and DNS Server roles are installed. After the installation is complete, the server is promoted to a Domain Controller by creating a new forest with a domain name (for example, company.local). During the promotion process, a Directory Services Restore Mode (DSRM) password is configured, and the server restarts. After rebooting, the server functions as the Domain Controller and DNS server for the network.

2. Directory Structure: Create 3 Organizational Units (OUs), 5 User Accounts, and 2 Security Groups via Active Directory Administrative Center.

Using the Active Directory Administrative Center, three Organizational Units (OUs) are created, such as HR, Sales, and IT. Inside these OUs, five user accounts are created with usernames, passwords, and login details. Two security groups, such as HR_Group and IT_Group, are also created to simplify permission management. Users are added to their respective groups based on their department. This logical structure makes administration easier and allows Group Policies and permissions to be applied efficiently.

3. Policy Enforcement: Configure a GPO to enforce complex passwords and restrict Control Panel access for specific OUs.

A Group Policy Object (GPO) is created using the Group Policy Management Console (GPMC). Password policies are configured to enforce password complexity, minimum password length, and password expiration settings for domain users. Another policy is configured under User Configuration → Administrative Templates → Control Panel to prevent access to the Control Panel. The GPO is then linked to the required Organizational Unit (OU) so that only users in that OU receive the restriction. After running `gpupdate /force`, the policy is applied successfully.

4. Resource Management: Create a shared folder on an NTFS volume with granular NTFS and Sharing permissions for defined groups.

A folder is created on an NTFS-formatted drive and shared over the network. Sharing permissions are configured to allow access only to authorized security groups, while NTFS permissions are assigned with different access levels such as Read, Modify, or Full Control.

based on user roles. For example, the HR_Group may receive Modify permission while other users receive only Read permission. Since NTFS and Share permissions work together, the most restrictive permission is applied, ensuring secure access to sensitive files.

5. Network Automation: Authorize a DHCP scope to automate IP leasing and verify connectivity using ipconfig /renew on a client.

The DHCP Server role is installed and authorized in Active Directory. A new DHCP scope is created by specifying the IP address range, subnet mask, default gateway, DNS server, and lease duration. The scope is then activated so that client computers can automatically receive IP addresses. On the Windows client machine, the command `ipconfig /renew` is executed in Command Prompt to request a new IP address from the DHCP server. The assigned IP address can be verified using `ipconfig /all`, confirming that automatic IP allocation and network connectivity are working correctly.

Section B: Practical Application

1. Core Services: Install Windows Server and promote it to a Domain Controller with AD DS and DNS.

To configure the server, Windows Server 2022 is first installed on a virtual machine using VMware or VirtualBox. After installation, a static IP address is assigned to the server. Using Server Manager, the Active Directory Domain Services (AD DS) and DNS Server roles are installed. After the installation is complete, the server is promoted to a Domain Controller by creating a new forest with a domain name (for example, company.local). During the promotion process, a Directory Services Restore Mode (DSRM) password is configured, and the server restarts. After rebooting, the server functions as the Domain Controller and DNS server for the network.

2. Directory Structure: Create 3 Organizational Units (OUs), 5 User Accounts, and 2 Security Groups via Active Directory Administrative Center.

Using the Active Directory Administrative Center, three Organizational Units (OUs) are created, such as HR, Sales, and IT. Inside these OUs, five user accounts are created with usernames, passwords, and login details. Two security groups, such as HR_Group and IT_Group, are also created to simplify permission management. Users are added to their respective groups based on their department. This logical structure makes administration easier and allows Group Policies and permissions to be applied efficiently.

3. Policy Enforcement: Configure a GPO to enforce complex passwords and restrict Control Panel access for specific OUs.

A Group Policy Object (GPO) is created using the Group Policy Management Console (GPMC). Password policies are configured to enforce password complexity, minimum

password length, and password expiration settings for domain users. Another policy is configured under User Configuration → Administrative Templates → Control Panel to prevent access to the Control Panel. The GPO is then linked to the required Organizational Unit (OU) so that only users in that OU receive the restriction. After running gpupdate /force, the policy is applied successfully.

4. Resource Management: Create a shared folder on an NTFS volume with granular NTFS and Sharing permissions for defined groups.

A folder is created on an NTFS-formatted drive and shared over the network. Sharing permissions are configured to allow access only to authorized security groups, while NTFS permissions are assigned with different access levels such as Read, Modify, or Full Control based on user roles. For example, the HR_Group may receive Modify permission while other users receive only Read permission. Since NTFS and Share permissions work together, the most restrictive permission is applied, ensuring secure access to sensitive files.

5. Network Automation: Authorize a DHCP scope to automate IP leasing and verify connectivity using ipconfig /renew on a client.

The DHCP Server role is installed and authorized in Active Directory. A new DHCP scope is created by specifying the IP address range, subnet mask, default gateway, DNS server, and lease duration. The scope is then activated so that client computers can automatically receive IP addresses. On the Windows client machine, the command ipconfig /renew is executed in Command Prompt to request a new IP address from the DHCP server. The assigned IP address can be verified using ipconfig /all, confirming that automatic IP allocation and network connectivity are working correctly.

Section C: Integrated Case Study

1. Analysis: Identify three specific GPO settings or NTFS conflicts that could cause users to log in successfully but lose access to shared drives.

Users may be able to log in successfully but lose access to shared folders due to the following reasons:

- **Incorrect NTFS Permissions:** A new GPO or manual change may have removed the required Read or Modify permissions from the users or security groups.
- **Restricted Group Policy Settings:** A GPO may have changed user rights or applied security filtering that prevents users from accessing network shares.
- **GPO Replication Failure:** If all Domain Controllers have not received the latest GPO, users may receive different policies depending on which Domain Controller authenticates them, causing inconsistent access to shared resources.

These issues can result in users authenticating successfully but receiving "Access Denied" when accessing shared folders.

2. Troubleshooting: Detail the step-by-step process of using the Resultant Set of Policy (RSOP) or gpresult tool to diagnose this issue.

To diagnose the issue:

1. Log in to the affected user's computer.
2. Open Command Prompt as Administrator.
3. Run the command:

```
gpresult /r
```

This displays the Group Policies currently applied to the user and computer.

4. For a detailed report, run:

```
gpresult /h C:\GPReport.html
```

5. Open the generated HTML report and verify which GPOs have been applied.
6. Alternatively, open Run, type rsop.msc, and press Enter.
7. Review the applied security settings, user rights, and administrative templates.
8. Compare the applied policies with the expected configuration to identify any conflicting GPO.

This process helps determine whether the issue is caused by an incorrect or missing Group Policy.

3. Tooling: List the specific CLI commands and GUI consoles you would use to verify if the file server is correctly seeing the user's updated security tokens.

CLI Commands

- `whoami /groups` – Displays the security groups assigned to the logged-in user.
- `gpresult /r` – Shows the applied Group Policies.
- `klist` – Displays Kerberos tickets.
- `gpupdate /force` – Refreshes Group Policies.
- `whoami /all` – Displays complete user security information.

GUI Consoles

- Active Directory Users and Computers (ADUC) – Verify user and group membership.
- Group Policy Management Console (GPMC) – Check GPO configuration and links.
- Resultant Set of Policy (rsop.msc) – View effective policies.
- File Explorer → Folder Properties → Security Tab – Verify NTFS permissions.
- Computer Management → Shared Folders – View shared folders and active sessions.

These tools help confirm whether the user's security token contains the correct group memberships and permissions.

4. Resolution: Propose a final fix to restore access and a preventive strategy, such as A-G-DL-P, to ensure future policy changes do not break permissions.

To restore access, first verify that the affected users are members of the correct Active Directory security groups. Check both NTFS and Shared Folder permissions to ensure the required groups have the appropriate access rights. Run `gpupdate /force` on client computers and `repadmin /syncall` on Domain Controllers to synchronize Group Policies. If required, restart the client computer so that updated security tokens are generated.

For future prevention, implement the A-G-DL-P model:

- A – Accounts: User accounts
- G – Global Groups: Users are added to Global Groups.
- DL – Domain Local Groups: Global Groups are added to Domain Local Groups.
- P – Permissions: Permissions are assigned only to Domain Local Groups.

This method simplifies permission management and reduces the chance of permission errors after GPO changes.

5. Validation: Explain how to use the net share and icacls commands to confirm that the folder permissions match the intended business logic.

The `net share` command is used to display all shared folders on the server and verify that the required folder is shared correctly.

Example:

```
net share
```

The `icacls` command displays the NTFS permissions assigned to a folder.

Example:

```
icacls D:\ProjectFiles
```

The output shows which users and groups have Read, Modify, or Full Control permissions. Compare these permissions with the organization's access requirements to ensure only authorized groups have access. If the displayed permissions match the intended business logic, the shared folder is correctly configured. Otherwise, modify the permissions to restore secure and appropriate access.